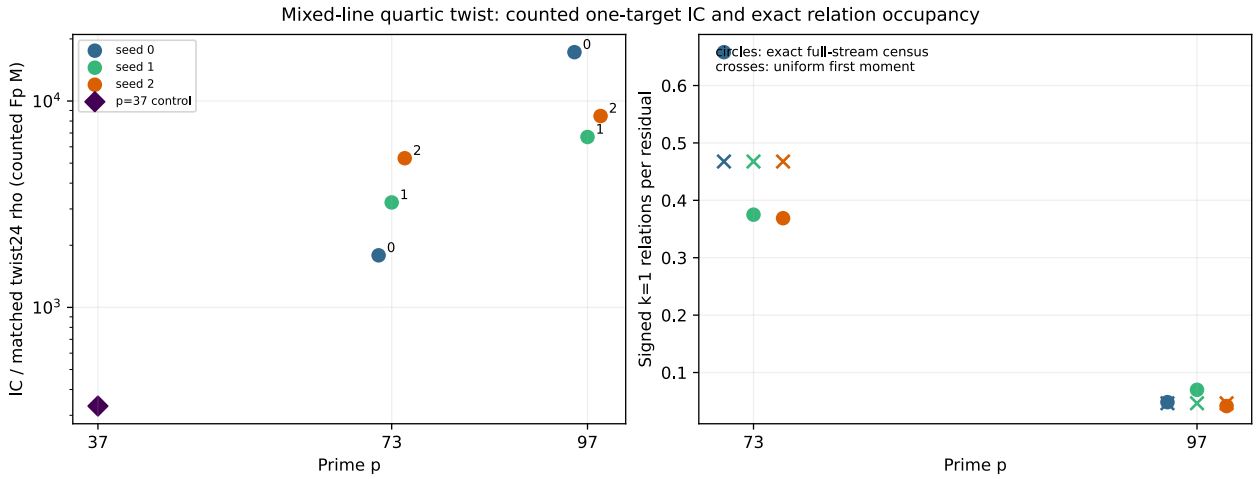
Twist symmetry in mixed-component polynomial index calculus

23 October 2026. Public synthetic one-target quartic-twist workloads at $p = 73$ and $p = 97$, each with $Q = [271]G$ and three frozen independent seeds. Protocol, raw records, independent audits, per-run PDFs and source links are in `K1_SCALING_REPORT.md`, `K1_SCALE_RUNS.md`, and `K1_CENSUS_RUNS.md`.

Mechanism

The transported Frobenius and cubic automorphism give a signed 24-action. Each tested nonzero abscissa orbit has size 12. A mixed base with B_0 points on $L_0 = F_p$ and B_2 on $L_2 = t^2 F_p$ has $\frac{B_0+B_2}{12}$ folded log columns: eight for the $48 + 48$ base at $p = 73$, six for the $36 + 36$ base at $p = 97$. Restricting to three L_0 factors and one L_2 factor permits a signed L_2 peel followed by a symmetrised S_4 solve in three L_0 abscissae. The 24 images of a folded relation row are scalar multiples of that row, so symmetry does not provide 24 new independent equations. This was checked on all 42 selected rows under all 24 actions: 1,008 exact row/RHS equalities and 12,096 factor-image closure checks.



Six one-target polynomial runs

p/seed	Queries	Exact/poly	Rank	IC Fp M	IC/rho Fp M
73/0	15	8/8	8/8	888,203,272	1,788
73/1	20	11/9	8/8	1,183,617,228	3,225
73/2	43	15/15	8/8	2,542,727,892	5,283
97/0	310	16/16	6/6	13,839,508,536	17,249
97/1	75	6/6	6/6	3,349,559,070	6,695
97/2	150	6/6	6/6	6,696,848,851	8,461

Every target was independently verified as 271. Five processed-prefix signed sets matched exactly. The $p = 73$, seed-one S_4 collector missed two relations at one failed system (query index 18), but still reached anchored rank eight and recovered the target. It remains an explicit partial relation-finder result. All first/A/A non-timing records match.

Complete frozen-stream occupancy

There are exactly $16\binom{B_0}{3}B_2$ signed distinct-index $k = 1$ tuples, giving expected set size $\lambda_1 = 16\binom{B_0}{3}\frac{B_2}{r}$ for a uniformly random residual. This first moment does not imply independent hits or guaranteed matrix rank. A point-keyed group census and independent BSGS-derived scalar-keyed auditor matched all 1,856 per-query complete-set digests, including post-stop queries.

p/seed	Stream	Exact k=1	Zero queries	Uniform total
73/0	117	77	57	54.71
73/1	120	45	80	56.11
73/2	122	45	88	57.05
97/0	558	27	535	25.92
97/1	458	32	427	21.27
97/2	481	20	461	22.34

At $p = 97$, 1,423 of 1,497 residuals have no $k = 1$ relation. The polynomial producer agrees with the exact oracle on every processed $p = 97$ query, so low physical occupancy explains these long collection paths. A one-count mutation in the census was rejected by the scalar auditor at query zero.

Claim boundary

The displayed ratios compare counted field multiplications only. Native hash/tree and allocator CPU, requested peak bytes, calibrated group/field bands, isolated interleaved timing, and same-input msolve/Magma results remain missing. They are not whole-method speed ratios or an admitted leaderboard result. The p=37 control had a counted ratio of 332.6 on a different base. This finite three-prime panel does not establish an asymptotic trend or a deployed-curve claim. The complete Markdown report names the next failure repair, base-shape comparison, and pricing gates.

Follow-up (23 October): native base shape and symmetry reuse

The separately registered 21-case native orbit-complete thinning grid ran every frozen prefix choice at p=73/97. Every case reaches full anchored rank and independently recovers 271; 17 relation sets are exact, while four p=73 seed-one choices retain the same two missing relations from one failed S4 system. The largest observed proper-subset same-seed full/candidate counted field ratios are 3.653 at p=73 seed two, one orbit, and 3.575 at p=97 seed zero, two orbits. The p=97 seed-one one-orbit choice regresses to 0.745. These are arithmetic-path observations, not a priced S or selected recipe.

The separate exact peeled-target orbit test finds two same-residual collisions among 142,248 finite signed L2 peels. Identity, cubic, transported Frobenius and combined caches find 20, 107, 243 and 790 total duplicates across six separate streams. The subgroup ablation isolates the small incremental twist cache opportunity on these instances. Source-linked full tables, figures and independent checks are in `K1_THIN_NATIVE_REPORT.md` and `K1_PEELED_ORBIT_REUSE.md`.

